Aiden Mathews

US Citizen | Aldie, VA | 2023amathews@gmail.com | linkedin.com/in/aiden-mathews | aidenmathews.vercel.app

Education

Virginia Tech — Blacksburg, VA

B.S. in Computer Science (GPA: 3.72) — *Expected May 2028*

- Relevant Coursework: Computer Systems, Introduction To AI, Machine Learning, Data Structures & Algorithms, Computer Organization, Discrete Mathematics, Cloud Software Development, Probability & Statistics

Experience

Freddie Mac — McLean, VA

Software Engineering Intern — *May 2026 — August 2026*

- Architected and built AI agent skills and a Node.js MCP server for semantic grouping of customer enhancement requests using LLM-based categorization, reducing manual triage time by 40% across hundreds of incoming tickets.
- Developed a Spring Boot backend with REST APIs for receiving, persisting, and managing customer enhancement requests in MongoDB, supporting the internal enhancements portal used by 50+ internal stakeholders.
- Built a Chatbot UI using Angular to surface categorized enhancement request insights, cutting stakeholder review time per request by 30% through conversational search and filtering.
- Validated API endpoints locally using Bruno and verified build readiness through Jenkins CI pipelines prior to deployment, reducing environment-related deployment failures by 25%.

Juume AI — Reston, VA

Junior Fullstack Engineering Intern — *June 2025 — June 2026*

- Designed and deployed a Go (Gin)-based Analytics REST API in an Agile environment, exposing endpoints for gateway usage, token consumption, and performance metrics, improving observability for internal teams.
- Integrated Temporal workflows, DuckDB, Postgres, and GCS for log analytics and PII detection, implementing prepared statements and cron jobs to reduce API latency by 35% and detect 93% of PII.
- Provisioned a Langfuse observability stack on Google Cloud using Terraform, integrating private VPC networking, Redis (Cloud Memorystore), Cloud SQL PostgreSQL, ClickHouse (trace/analytics storage), and GCS with lifecycle policies.
- Modernized an AI Gateway by refactoring the Go control plane with embedded PostgreSQL and resolving IAM token refresh race conditions, reducing gateway crash incidents by 90% and enabling zero-downtime deployments on Cloud Run.
- Automated secure secret management using Google Secret Manager and scoped IAM roles, strengthening compliance.

Bunchful Enterprise — New York, NY

Software Engineering Intern — *September 2024 — January 2025*

- Led migration of Bunchful.news from WordPress to React, implementing React Router and Firebase for real-time updates, improving site load speed by 40%.
- Developed responsive UI components with React and CSS, achieving 100% cross-browser/device compatibility and improving mobile page speed scores by 22 points.
- Wrote comprehensive tests using Jest and React Testing Library, reducing regression bugs and boosting test coverage to 85%.
- Streamlined deployments via Netlify and Vercel, reducing deployment time by 50%.
- Leveraged Axios for API integration to dynamically fetch and display content from Firebase, reducing content load time by 20%.

Projects

Agentic Decision Evaluation Harness | *TypeSafe Jev, Claude API, Python, MCP* — *September 2026*

- Architected an MCP-based evaluation harness comparing TypeSafe's Jev decision model against Claude on infrastructure tool-routing and risk-gating decisions (Terraform, kubectl) using identical typed output schemas.
- Measuring tool-selection accuracy, hallucination rate, calibration error (Brier score), latency, and cost per decision, including a confidence-threshold hybrid strategy that escalates low-confidence Jev outputs to Claude.

Smart Legal Document Analyzer | *React, FastAPI, OpenAI API* — *July 2025*

- Developed a legal contract analyzer with LLM-based extraction, PDF parsing, and prompt engineering, deployed on a React frontend with a FastAPI backend (Vercel serverless), improving user comprehension by 30% and reducing load times by 35%.

Technical Skills

Languages: JavaScript, Python, Java, Go, C++, C#, SQL, HTML/CSS
Frameworks: React, Angular, Spring Boot, Node.js, FastAPI, Gin, Tailwind
Databases: MongoDB, PostgreSQL, DuckDB, ClickHouse, Firebase
Cloud & Tools: GCP, AWS, Terraform, Docker, Redis, Git, Vercel, OpenAI API, MCP, Langfuse, Agile/Scrum, CI/CD